MICROSOFT AI TOOLKIT · RAGAS · FASTAPI

Evaluating AI Agents That Actually Work

RAG Quality Metrics for E-Commerce & Custom Docs
with RAGAS, FastAPI & Microsoft AI Toolkit

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RAGAS

FastAPI

AI Toolkit

Python

RAG

Building AI You Can Trust — Not Just AI You Can Demo

THE PROBLEM

Building AI is easy. Knowing it works is hard.

Looks Good in Demo

The bot answers 10 hand-picked questions perfectly. But what about the 500 edge cases your users will actually ask?

Silent Hallucinations

The AI confidently states wrong information drawn from outside the knowledge base. You won't catch it without ground-truth comparison.

No Regression Signal

You change the system prompt or swap the model. Did quality improve or regress? Without evals, you're flying blind.

Evaluations turn gut feelings into measurable, repeatable quality signals.

Today's Agenda

What We'll Build

01 **The Evaluation Problem**

Why 'it works on my machine' isn't enough for production AI

02 **CustomerBot — ShopEval Demo**

RAG pipeline for e-commerce + live RAGAS metric scoring

03 **Bring Your Own Docs — RAGEval**

Upload PDFs/DOCX/TXT → auto-generate questions → evaluate

04 **AI Toolkit Evaluation Suite**

Bulk Run, Evaluation Tab, Copilot Code Gen — inside VS Code

05 **Key Takeaways & Next Steps**

From demo to production: vector DBs, real RAGAS, CI/CD evals

RAGAS FRAMEWORK

4 Metrics That Tell You If Your RAG Actually Works

92%

Faithfulness

Are all claims in the answer grounded in the retrieved context?
Detects hallucination — the AI's cardinal sin.

87%

Answer Relevancy

Does the answer actually address what was asked?
Detects off-topic, watered-down, or evasive responses.

78%

Context Precision

Of retrieved chunks, what fraction were truly relevant?
Measures retrieval signal-to-noise ratio.

83%

Context Recall

Was all ground-truth information present in context?
Measures retrieval coverage and completeness.

RAGAS Score = (Faithfulness + Answer Relevancy + Context Precision + Context Recall) ÷ 4 · Pass Threshold ≥ 0.70

RAGAS · Retrieval Augmented Generation Assessment

How Do You Know Your RAG Actually Works?

RAGAS is an open-source framework that uses an LLM-as-judge to score your RAG pipeline across 4 dimensions — turning gut feelings into measurable quality signals.

Question — Retrieve — Generate — Evaluate

How RAGAS Scores

1. Sends question + answer + context to an LLM judge
2. LLM decomposes claims and checks each one
3. Returns a 0–1 score per metric
4. No manual grading — fully automated

92%

Faithfulness

Is every claim grounded in the retrieved context?

Detects hallucination — the AI's cardinal sin

87%

Answer Relevancy

Does the answer address what was actually asked?

Catches off-topic, evasive or padded replies

78%

Context Precision

Were the retrieved chunks truly relevant?

Measures retrieval signal-to-noise ratio

83%

Context Recall

Was all necessary info present in the context?

Measures retrieval coverage & completeness

RAGAS Score = (Faithfulness + Answer Relevancy + Context Precision + Context Recall) ÷ 4 · Pass Threshold ≥ 0.70

DEMO 1

ShopEval — CustomerBot

E-Commerce RAG Agent with Live RAGAS Evaluation Dashboard

1 Ask

Type a question

Answer streams in

RAGAS scores animate

2 Evaluate

Pick from test set

Bot vs Ground Truth

Full metric table

3 Results

Run all 8 tests

Progress bar

Per-test breakdown

📚 Knowledge Base

8 e-commerce policy docs · Returns, Shipping, Payments, Warranty, Loyalty

🧪 Test Dataset

8 ground-truth Q&A pairs · Pre-loaded · RAGAS scored against threshold 0.70

HOW IT WORKS

The RAG Pipeline — From Question to Evaluated Answer

① User Question

Natural language query from user

② Retrieve

Top-K chunks from knowledge base (vector search)

③ Generate

LLM synthesises answer from retrieved context

④ Evaluate

RAGAS scores answer against ground truth

⑤ Pass / Fail

Score ≥ 0.70 = Pass · Score < 0.70 = Fail

RAGAS Evaluation Layer

- Faithfulness
- Relevancy
- Precision
- Recall

DEMO 2

RAGEval — Bring Your Own Docs

Upload any PDF, DOCX, or TXT → Auto-generate test questions → Evaluate with RAGAS

Step 1 · Upload

Drag & drop PDF, DOCX, or TXT
Text is parsed and chunked automatically into ~400-word segments

Step 2 · Questions

AI auto-generates Q&A pairs from your document content
Edit, delete, or add manually

Step 3 · Ask

Chat with your documents
Answer streams with typewriter
RAGAS scores animate in real-time

Step 4 · Results

Run all test cases at once
Bot answer vs ground truth
Expand each result for full breakdown

PDF | DOCX | TXT

supports any domain — HR policies, legal docs, product manuals, research papers, and more

Native Evaluation Features — Inside VS Code

Agent Builder + Evaluation Tab

Build your agent prompt, switch to the Evaluation tab, load a JSONL dataset, and manually rate responses with thumbs up/down. Perfect for quick qualitative review.

Bulk Run

Test your agent or prompt against multiple inputs in a single batch operation. Generates all outputs at once — then feed them to the Evaluation step for scoring.

Standalone Evaluation View

TOOLS → Evaluation → Create Evaluation. Choose evaluator (Faithfulness, Relevance, Coherence, Task Adherence), set judging model, load JSONL, compare runs.

Copilot Eval Code Gen

Ask GitHub Copilot: 'Create an evaluation for my RAG app'. It reviews your app, generates synthetic test questions, runs batch eval, and writes Azure AI Eval SDK code.

JSONL format for AI Toolkit: { "query": "...", "response": "", "ground_truth": "..." } — response field auto-filled by Bulk Run

TECH STACK

Everything You Need — No Cloud Required

Microsoft AI Toolkit

Local model inference & evaluation suite inside VS Code

RAGAS Framework

Open-source RAG evaluation library — LLM-as-judge metrics

FastAPI

REST API layer — RAG pipeline, evaluation endpoints, file upload

GitHub Copilot

Evaluation code generation, prompt iteration, UI scaffolding

Python

pypdf · python-docx · pydantic · uvicorn

HTML / CSS / JS

Single-file dashboard — no framework, open directly in browser

Replace the Demo Stubs With the Real Thing

Demo (Today)	Production Replace
Keyword overlap retrieval	→ FAISS / Chroma / Qdrant + sentence-transformers embeddings
Heuristic answer generation	→ Real LLM call — Azure OpenAI GPT-4o / AI Toolkit local model
Simulated RAGAS metrics	→ pip install ragas → from ragas import evaluate + LLM judge
Heuristic question generation	→ LLM generates diverse, adversarial Q&A pairs automatically
Manual evaluation run	→ CI/CD pipeline: eval runs on every PR — block if RAGAS < 0.70

KEY TAKEAWAYS

1

Evals are not optional.

They are the difference between shipping AI and gambling with AI.

2

RAGAS gives you 4 orthogonal lenses.

Faithfulness · Relevancy · Precision · Recall — each catches a different failure mode.

3

Ground truth quality drives eval quality.

Garbage in, garbage out. Time spent on ground truth is time saved in debugging.

4

AI Toolkit brings evals inside VS Code.

Bulk Run + Evaluation Tab + Copilot Code Gen — no context switching required.

5

Start simple, graduate to production.

Keyword retrieval → vector search. Simulated metrics → real ragas library. Manual → CI/CD.

SESSION FLOW

60-Minute Demo Breakdown

1 **Why Evals Matter** — 17 min

The cost of hallucination. Gap between 'demo works' and 'production works'. Stakes in e-commerce.

2 **ShopEval Live Demo** — 17 min

Ask tab → typewriter answer → RAGAS scores animate. Show Pass vs Fail by tweaking retrieval.

3 **Full Suite + Results** — 17 min

Run all 8 tests. Progress bar. Expand per-test breakdown. Bot answer vs ground truth side-by-side.

4 **RAGEval — Your Own Docs** — 25 min

Upload a real document. Auto-generate questions. Edit ground truth. Ask the bot. Run evaluation.

5 **AI Toolkit Evaluation Suite** — 17 min

Bulk Run → Evaluation Tab → Copilot Code Gen. Show JSONL dataset. Compare two prompt versions.

6 **Takeaways & Next Steps** — 8 min

Replace simulated RAGAS with real library. Vector DB for retrieval. CI/CD eval pipelines.

Thank You

Now let's build AI you can actually trust.

RAGAS Documentation:	docs.ragas.io
AI Toolkit for VS Code:	aka.ms/ai-toolkit
ShopEval + RAGEval Code:	https://github.com/Azure-Samples/AI_Toolkit_Samples
Azure AI Evaluation SDK:	https://learn.microsoft.com/en-us/azure/foundry-classic/how-to/develop/evaluate-sdk